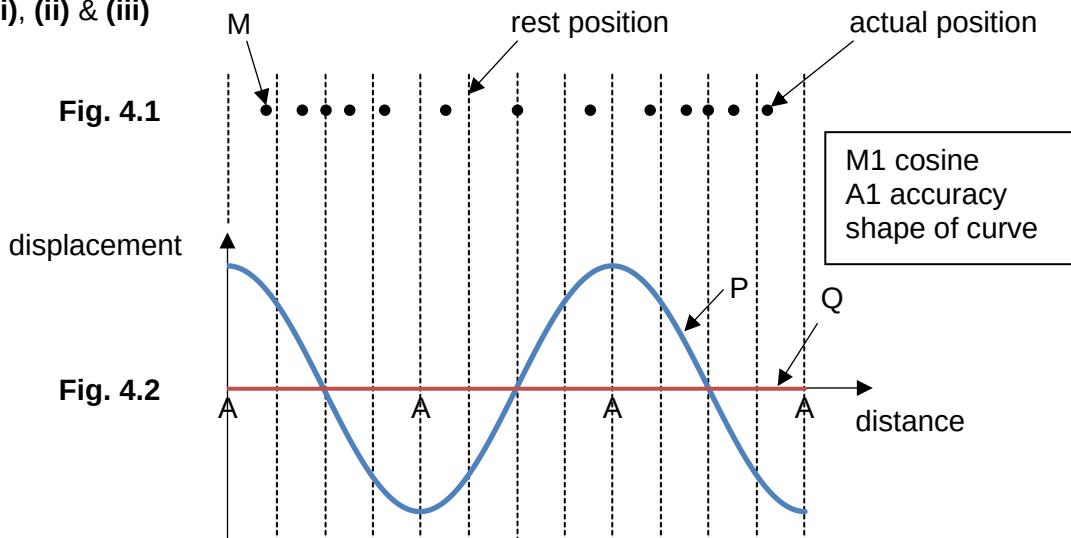


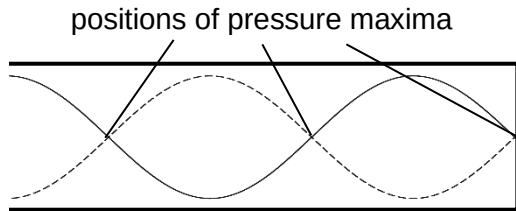
4 (a) (i), (ii) & (iii)



- (b) (i) The incident wave reflects off the wall of the pipe at the end, producing a wave of the same amplitude, wavelength and frequency travelling in the opposite direction. The reflected wave interferes with subsequent incident wave, to produce nodes and antinodes at different positions of the pipe, depending on the path difference between the two waves.

4

(ii)



The positions of maxima correspond to pressure antinodes (displacement nodes). As shown in the picture above, three positions of maxima imply that the length of the pipe is equal to $5/4$ wavelengths.

$$\frac{5}{4}\lambda = 0.40 \Rightarrow \lambda = 0.32 \text{ m}$$

(iii) $v = f\lambda = 1060 \times 0.32 = 339 \text{ m s}^{-1}$

5 (a) $e.m.f. = 2.0 \text{ V}$